

Consider the following class named **DataUsagePlan**.

Attributes:

- **numOfPlans**: a static variable. The total number of created plans.
- **customerName**: the name of the customer.
- **nationalID**: the National ID of the customer.
- **planID**: a unique ID for each plan.
- **planType**: the type of the data plan:
 - Light ('L'),
 - Standard ('S'),
 - Unlimited ('U').
- **usedGB**: the amount of used data in GB.

DataUsagePlan
- <u>numOfPlans</u> : int - <u>customerName</u> : String - <u>nationalID</u> : String - <u>planID</u> : String - <u>planType</u> : char - <u>usedGB</u> : int
+ DataUsagePlan(name:String, nationalID:String, type:char, usedGB:int) - generatePlanID(): String - isValidPlanType(char type) + calculateTotalCost(): double + displayInfo(): void + setters & getters

Methods:

DataUsagePlan(name:String, nationalID:String, type:char, usedGB:int)	Parameterized constructor. Use the argument values for customerName , nationalID , planType , and usedGB . Then generate a value for planID . Then increase <u>numOfPlans</u> by 1.
generatePlanID()	Generates a unique plan ID in the following format: • First character of nationalID • Last character of nationalID • Followed by the planType • Followed by the current numOfPlans Example: If the National ID is "1234567890", the plan type is 'L', and numOfPlans is 4, the generated ID will be: 10L4 Assume that the National ID is not empty.
calculateTotalCost()	Calculates and returns the total cost of the data plan according to the following rules: • Light ('L') ◦ If usedGB ≤ 10, the total cost is 50 SAR ◦ Otherwise, the total cost is: 50 + (usedGB - 10) × 5

	• Standard ('S') ○ If usedGB ≤ 50, the total cost is 100 SAR ○ Otherwise, the total cost is: $100 + (\text{usedGB} - 50) \times 3$ • Unlimited ('U') ○ The total cost is always 200 SAR
<u>isValidPlanType(char type)</u>	Checks whether the plan type is valid (either 'L', 'S', or 'U')
<u>displayInfo()</u>	Displays the plan information, including customer name, National ID, plan ID, plan type, used GB, and total cost.

Question)

- Implement class **DataUsagePlan**.
- Implement another class called **DataUsagePlanDemo** with:
 - **main** method should:
 - a. Read user input for customer name, National ID, plan type, and used GB.
 - b. Create an object using the parameterized constructor with user inputs -use **isValidPlanType()** to validate the plan type-.
 - c. Display all plan information, including the calculated total cost.
 - d. Repeat steps (a–c) until the user enters "exit" as the customer name.

Sample run

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Enter customer name: Ali
Enter National ID: 1234567890
Enter plan type ('L', 'S', or 'U'): L
Enter used GB: 8
Name: Ali, National ID: 1234567890, ID: 10L0, Type: L, Used GB: 8, Total
cost: 50.0 SAR
=====

Enter customer name: Sara
Enter National ID: 9876543211
Enter plan type ('L', 'S', or 'U'): X
ERROR: Incorrect plan type. Type can be 'L', 'S', or 'U' only
Enter plan type ('L', 'S', or 'U'): L
Enter used GB: 10
Name: Sara, National ID: 9876543211, ID: 91L1, Type: L, Used GB: 10, Total
cost: 50.0 SAR
=====

Enter customer name: Omar
Enter National ID: 1122334455
Enter plan type ('L', 'S', or 'U'): L

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Enter used GB: 14
Name: Omar, National ID: 1122334455, ID: 15L2, Type: L, Used GB: 14, Total
cost: 70.0 SAR
=====

Enter customer name: Huda
Enter National ID: 5566778899
Enter plan type ('L', 'S', or 'U'): S
Enter used GB: 50
Name: Huda, National ID: 5566778899, ID: 59S3, Type: S, Used GB: 50, Total
cost: 100.0 SAR
=====

Enter customer name: Nasser
Enter National ID: 6677889900
Enter plan type ('L', 'S', or 'U'): S
Enter used GB: 65
Name: Nasser, National ID: 6677889900, ID: 60S4, Type: S, Used GB: 65,
Total cost: 145.0 SAR
=====

Enter customer name: Reem
Enter National ID: 1029384756
Enter plan type ('L', 'S', or 'U'): U
Enter used GB: 25
Name: Reem, National ID: 1029384756, ID: 16U5, Type: U, Used GB: 25, Total
cost: 200.0 SAR
=====

Enter customer name: Majed
Enter National ID: 5647382910
Enter plan type ('L', 'S', or 'U'): U
Enter used GB: 120
Name: Majed, National ID: 5647382910, ID: 50U6, Type: U, Used GB: 120,
Total cost: 200.0 SAR
=====

Enter customer name: exit
```